

Assignment - 3

1. a. Thermal Voltage & Diode Current

Temperature in kelvin $T_K = 25 + 273 = 298 \text{ K}$

V_T :

$$V_T = \frac{(1.38 \times 10^{-23})(298)}{1.602 \times 10^{-19}}$$

$$V_T = 0.02567 \text{ V} / 25.67 \text{ mV}$$

b. Find the diode current, we can use the given value

$$I_S = 40 \text{ nA}, n=2, V_D = 0.5 \text{ V}$$

$$\text{Equation - } I_D = I_S \left(e^{\frac{V_D n}{V_T}} - 1 \right)$$

$$\text{Calculate: } I_D = (40 \times 10^{-9}) \times \left(e^{\frac{0.5}{2 \times 0.02567}} - 1 \right)$$

$$I_D = (40 \times 10^{-9}) \times (e^{9.739} - 1)$$

$$I_D = 0.000678 \text{ A or } 0.678 \text{ mA}$$

2. Apx. Characteristics in the model, a forward-biased Silicon diode has a voltage drop of 0.7V

- V_D = The voltage across the diode is simply 0.7V
- V_R = The remaining voltage from the 30V source the 1.5 k Ω resistor.

$$V_R = 30 - 0.7 = 29.3V$$

- I_D = Using Ohm's law ($I = \frac{V}{R}$) on the matter
$$I_D = \frac{29.3}{1500} = 0.01953 \text{ A}$$

b. The diode acts as a perfect switch with a 0V drop

- V_D = The voltage across the diode is 0V
- V_R = The entire 30V drops across the resistor
$$V_R = 30V$$

$$I_D = I_R = \frac{30}{1500} = 0.02A \text{ (20mA)}$$

c. Yes, the result suggests the ideal model provides a very good approximation in this scenario. Because the applied source voltage (30V) is significantly larger than the diode's built-in-drop (0.7V) the difference in calculated current is minimal.

Q.3

a. The diode is connected to the negative terminal and is reverse-biased, it acts as an open circuit

$$I = 0A$$

b. The 20V battery has its positive terminal connected to ground, making the top node -20V. The diode's anodes are connected to the ground (0V) through resistors. Since current flows from higher to lower potential (0V to -20V), both diodes are forward-biased

Left Branch Current (I_1)

$$0 - 0.7 - I_1(10) = -20 \rightarrow 10I_1 = 19.3$$

$$\rightarrow I_1 = 1.93A$$

Middle Branch Current (I_2)

$$0 - 0.7 - I_2(20) = -20 \rightarrow 20I_2 = 19.3$$

$$\rightarrow I_2 = 0.965A$$

Total Current (I) =

$$I = I_1 + I_2 = 1.93 + 0.965 = 2.895A$$

The current I flows entirely through the outer loop with the 10V battery & 10Ω . $\therefore I = 10/10 = 1A$

Q.4.

Circuit a

The silicon diode (0.7V drop) and GaAs diode (1.2V drop) are in parallel. The silicon diode will turn on first, clamping the voltage across the parallel block to 0.7V. The GaAs diode will remain off.

$$V_o = V_o = 1 - 0.7 = 0.3V$$

$$I = I = \frac{V_R}{R} = \frac{0.3}{1000} = 0.0003A \text{ or } 0.3mA$$

Circuit B

The current flows through the top Si diode and then splits through 2 parallel Si diodes. The total voltage drop across the diodes is 1.4V

$$V_o = V_o = 16 - 1.4 = 14.6V$$

$$V_{\text{resistor}} = 14.6 - (-4) = 18.6V$$

$$I = \frac{18.6}{4700}$$

$$= 0.003957A \text{ or } 3.957mA$$